Gap-Stable Spectral Packet Enclosures
Projector Certification, Polar Frame Alignment, and a Sharp Crossing Wall

Prime Dynamics Theory Program

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Abstract

RH-140 converts validated source-state residuals into operator-norm balls for trace-normalized Gram snapshots. We determine exactly when such a ball fixes a spectral packet. Let $\widehat{A}$ be Hermitian, let $\widehat{P}$ project onto its top r eigenvalues, let

$$g = \lambda_r(\widehat{A}) - \lambda_{r+1}(\widehat{A}), \qquad \|A - \widehat{A}\| \le \varepsilon.$$

If $g > 2\varepsilon$, then the top-r projector P of A is uniquely separated and

$$\|P - \widehat{P}\| \le \frac{\varepsilon}{g - \varepsilon}.$$

After canonical polar alignment, orthonormal packet frames satisfy

$$\|U - \widehat{U}O\| \le \sqrt{2 - 2\sqrt{1 - s^2}}, \qquad s = \|P - \widehat{P}\|.$$

The threshold $g = 2\varepsilon$ is sharp: at equality a two-dimensional example becomes degenerate, and beyond it the leading subspace may swap completely. No basis inside a multiple cluster is intrinsically stable, so the result certifies projectors and polar-aligned frames, not ordered eigenvectors.

The RH-140 universal balls certify four of ten rank-four anchor packets. The six failures occur at the three coarser scales and are rigorous gap walls for the available norm-ball information. In contrast, replacing the universal radius by the quadratic exact-SVD diagnostic makes all ten pass; the smallest margin is only 1.0213, at the coarsest right channel. Thus the next interface problem is sharply localized: validate the quadratic snapshot cancellation, including its small backward-error defect, before attempting a thresholded packet chain.

1 From a matrix ball to a packet

A small perturbation of a Hermitian matrix need not determine an eigenvector. Eigenvalues may cross, an eigenvalue may be multiple, and a numerical routine may rotate a basis arbitrarily inside a fixed invariant subspace. The coordinate-free object is the spectral projector onto an isolated cluster. RH-140 supplies a ball $\|A - \widehat{A}\| \le \varepsilon$; the missing datum is therefore the separation of the intended cluster.

For the source compression considered here, $\widehat{A}$ has rank at most four. Its top-four gap is the smallest retained normalized singular energy. This value can be extremely small at the coarser anchors even when total rank-four energy capture is excellent. Energy capture and packet identifiability are consequently different assertions.

2 Approximate-gap enclosure

We state the finite-dimensional theorem for a top cluster; the same argument applies to any separated interval.

Theorem 1 (Gap-certified packet). *Let $A, \widehat{A}$ be Hermitian matrices with $\|A - \widehat{A}\| \le \varepsilon$. Let $\widehat{P}$ be the top-r spectral projector of $\widehat{A}$, and define $g = \lambda_r(\widehat{A}) - \lambda_{r+1}(\widehat{A})$. If $g > 2\varepsilon$, then A has a positive top-r gap and its top-r projector P obeys*

$$\|P - \widehat{P}\| = \|\sin\Theta(P, \widehat{P})\| \le \frac{\varepsilon}{g - \varepsilon} < 1.$$

Proof. Weyl's inequality gives

$$\lambda_r(A) - \lambda_{r+1}(A) \ge g - 2\varepsilon > 0,$$

so the exact top cluster is well defined. Regard $\widehat{P}$ as an approximate invariant subspace for A. Its residual is $(I - \widehat{P})(A - \widehat{A})\widehat{P}$ and has norm at most ε. The selected approximate spectrum lies above $\lambda_r(\widehat{A})$, whereas the exact unwanted spectrum lies below $\lambda_{r+1}(A) \le \lambda_{r+1}(\widehat{A}) + \varepsilon$. Their separation is at least $g - \varepsilon$. The residual form of the Davis–Kahan sine theorem now gives the displayed bound [Davis and Kahan, 1970, Stewart and guang Sun, 1990]. □

The estimate is outward: only the approximate gap and the norm radius are needed. It is nontrivial exactly on the no-crossing side of the threshold.

3 Canonical frame alignment

Suppose U and $\widehat{U}$ are orthonormal frames for P and $\widehat{P}$. Their columns are not canonical. Let O be the polar factor solving the orthogonal Procrustes alignment between the frames.

Proposition 2 (Polar-aligned frame ball). *If $s = \|P - \widehat{P}\| < 1$, then*

$$\min_{O^*O=I} \|U - \widehat{U}O\| = 2\sin\left(\frac{\theta_{\max}}{2}\right) = \sqrt{2 - 2\sqrt{1 - s^2}},$$

where $\sin\theta_{\max} = s$. The minimizing O is the polar alignment.

Proof. Choose principal frames. The singular values of $\widehat{U}^*U$ are $\cos\theta_j$. Polar alignment pairs the principal vectors, and each two-frame difference has singular value $2\sin(\theta_j/2)$. Taking the maximum proves the formula. Arbitrary rotations within either frame cannot improve the largest principal pair [Bhatia, 1997]. □

This statement is the correct way to feed a packet into later frame-based calculations. An enclosure for ordered eigenvectors would be false whenever the selected cluster contains a repeated eigenvalue, even though the packet projector itself can be perfectly stable.

packet rank	universal stable	universal walls	quadratic diagnostic stable
one	7/10	3/10	10/10
two	7/10	3/10	10/10
four	4/10	6/10	10/10

Table 1: Finite packet gates. The last column is diagnostic, not interval validated.

4 The sharp two-radius wall

The condition in Theorem 1 cannot be weakened using only g and ε. Let

$$\widehat{A} = \begin{pmatrix} g/2 & 0 \\ 0 & -g/2 \end{pmatrix}, \qquad E = \begin{pmatrix} -\varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix}.$$

At $\varepsilon = g/2$, the perturbed matrix is zero and its leading projector is not unique. For $\varepsilon > g/2$, the leading coordinate swaps and the projector distance is one. Thus a failed $g > 2\varepsilon$ gate is not merely a weakness of Davis–Kahan: the same data permit complete packet loss.

Additional structure can still cross the wall. A correlated perturbation, an interval residual orthogonal to the selected singular spaces, or a direct resolvent enclosure may be much smaller than the ambient norm ball. Such information must be supplied explicitly.

5 Anchor audit

For each RH-140 rank-one, rank-two, and rank-four compressed state, we form the approximate normalized snapshot and measure its retained-to-omitted gap. The certified radius is inherited unchanged from the 160-bit Arb state residual. A 4,096-instance synthetic audit of stable random Hermitian perturbations has zero violations of Theorem 1.

For rank four, both channels at $\sigma = 0.02$ and 0.01 pass. The maximum certified projector radius among these four is 0.3991, and the corresponding polar-aligned frame radius is below 0.408. Both channels at $\sigma = 0.16, 0.08, 0.04$ fail, sometimes by many orders of magnitude. These are not declared approximately stable.

The ideal exact-SVD radius from RH-140 is q^2 rather than q. It crosses all ten rank-four gates. The smallest gap ratio $g/(2q^2) = 1.0213$ occurs for the coarsest right channel, so even that route has little spare room there. A validated orthogonality defect larger than about two percent of the ideal budget could close the gate. This narrow margin is useful negative information: it tells the next calculation exactly where outward rounding must be concentrated.

6 Consequence and claim boundary

RH-141 completes the abstract map

$$\text{snapshot ball} + \text{isolating gap} \longrightarrow \text{projector ball} \longrightarrow \text{polar-aligned frame ball.}$$

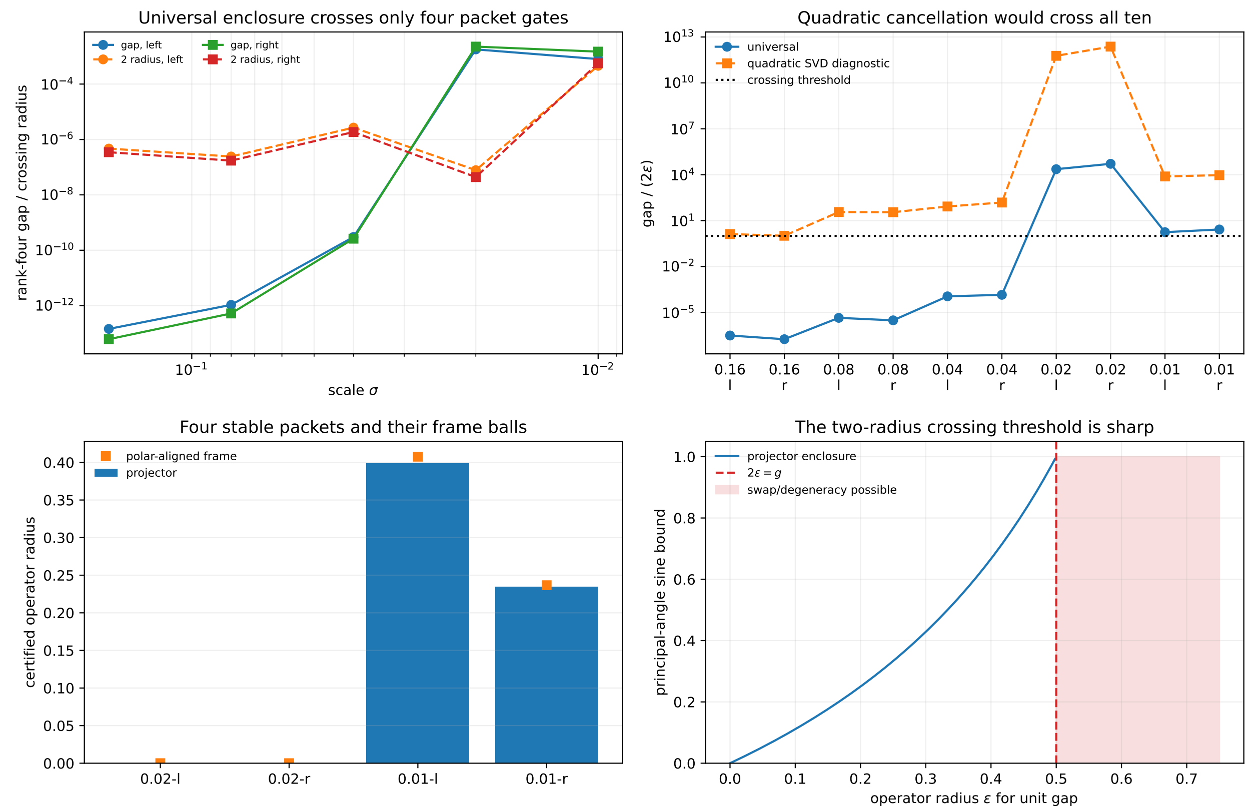

Figure 1: Rank-four gaps versus universal radii, universal and quadratic gap ratios, certified projector/frame balls, and the sharp crossing wall.

It also shows that the present universal state-to-snapshot bound is not sufficient for six of the intended rank-four anchors. The most economical next step is therefore not a larger thresholded simulation. It is a direct Arb enclosure of the normalized Gram difference that preserves the orthogonal-SVD quadratic cancellation and quantifies the float SVD defect.

We have not interval-validated that quadratic radius, certified all ten rank-four packets, enclosed any recursive threshold update, proved a uniform all-level gap, controlled the RH-139 viability tube or normalized-base liminf, established Stage A, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

References

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